CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 93-235

WASTE DISCHARGE REQUIREMENTS
FOR
CALIFORNIA DEPARTMENT OF CORRECTIONS
PLANNING AND CORRECTIONS DIVISION
FAIRMEAD PRISON COMPLEX
WASTEWATER TREATMENT FACILITY
MADERA COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. California Department of Corrections, Planning and Corrections Division, (hereafter Discharger) submitted a Report of Waste Discharge dated 10 April 1991, and a technical report dated March 1992, in support of an increase in discharge flows due to construction of a second multi-security level women's prison at the northwest corner of Avenue 24 and Road 22, near Fairmead, Madera County.

2. Waste Discharge Requirements Order No. 89-042, adopted by the Board on 31 March 1989, prescribes requirements for treatment of 0.685 mgd of domestic wastewater from the Central California Women's Facility (hereafter CCWF) and discharge to four evaporation/percolation ponds and 124 acres of agricultural fields.

3. Order No. 89-042 is inadequate due to an increase in flows and a change in disposal location.

4. The proposed second prison, north of the existing facility, is known as the California State Prison Madera County II (hereafter Madera County II). The prison will be a 1,980 bed prison, with the possibility of overcrowding to 190 percent of capacity, or 3,762 beds. The Discharger is also planning to construct a minimum security prison camp on 9 acres of existing wastewater reclamation fields.

5. The existing wastewater treatment facility (WWTF) provides pretreatment of influent wastewater with a grease trap, a mechanically-cleaned bar screen, and a grinder. Following pretreatment, wastewater is biologically treated in two 3.1-acre aerated lagoons followed by two 3.1-acre facultative ponds, each set adaptable to operation in parallel or series. The Discharger plans to install an additional bar screen and grinder, and construct one aerated lagoon and one facultative lagoon to facilitate flows generated by Madera County II and the minimum security prison camp. Design flow for the expanded WWTF is 1.454 mgd.

6. Wastewater from the WWTF discharges to four 6.1-acre effluent storage ponds for partial disposal by evaporation/percolation. The Discharger plans to construct one 6.5-acre pond and two 6.7-acre ponds for additional storage and disposal capacity.

7. During the growing season, wastewater will be pumped to 115 acres (existing) of agricultural land in the northwest quarter of Section 6, T10S, and 150 acres (proposed) in the western portion of Section 31, T9S, R17E, MDB&M, for reclamation by flood irrigation of alfalfa between the months of March and November. The properties are owned by the Discharger. Tail water captured during this period will be collected in a tail water ditch system, drain into a recirculation pond, and returned to the fields.

8. Sludge accumulating in the facultative ponds will be removed every two or three years when the ponds are taken alternately off-line for cleaning. Sludge will be spread on property controlled by the Discharger or will be landfilled.

9. The WWTF is in Section 6, T10S, R17E, MDB&M, as shown in Attachment A, which is attached hereto and part of this Order by reference. Surface water drainage is to the San Joaquin River. The site lies within the Berenda Creek hydrologic area (No. 545.30), as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.

10. The Board adopted a Water Quality Control Plan, Second Edition, for the San Joaquin River Basin (hereafter Basin Plan), which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.

11. The beneficial uses of the San Joaquin River (Sack Dam to Merced River) are agricultural supply; industrial process supply; water-contact and nonwater contact recreation; wildlife habitat; warm fresh water habitat, fish migration (warm and cold), and fish spawning (warm).

12. The site is predominantly surrounded by agricultural land.

13. The prison buildings and one existing residence are within 1/2 mile of the wastewater reclamation fields.

14. There are 10 irrigation wells and 3 domestic wells within the boundaries of the two prison facilities. Two additional domestic wells are proposed to meet water supply needs for Madera County II. Nine off-site irrigation wells and one domestic well are within 1/4 mile of prison boundaries. The domestic well is approximately 230 feet from

the proposed reclamation field. The Report of Waste Discharge includes statements in support of this separation based on factors including ground water depth, well construction, soil lithology, and berms along the field to prevent runoff. The accepted separation between wastewater disposal areas and domestic water supply wells is 500 feet.

15. A portion of the proposed wastewater reclamation field lies within the 100-year floodplain. Measures will be taken to ensure contact between reclamation water and surface water runoff from precipitation is minimized. These measures include not reclaiming wastewater from 1 December through February and having the disposal ponds empty on or about 1 October of each year. Surface water runoff from precipitation in the reclamation area will be collected in the tail water ditch system and recirculation pond. Should the pond become full, tail water pumps will be used to return the runoff to the fields for percolation.

16. Stormwater from the prison complexes is intercepted by ditches outside the perimeter security fence and is discharged to an on-site detention basin. The stormwater is drained from the basin into a trapezoidal channel and conveyed off-site to an existing drainageway.

17. The State Water Resources Control Board adopted Water Quality Order No. 92-12-DWQ on 18 September 1992 which amended Waste Discharge Requirements Water Quality Order No. 91-13-DWQ, NPDES General Permit No. CAS000001, for Discharges of Storm Water Associated with Industrial Activities (hereafter General Permit). Facilities requiring coverage by the General Permit are listed by category [by Standard Industrial Classification (SIC)] in 40 Code of Federal Regulations (CFR), Section 122.26 (b)(14)m and are required to submit a Notice of Intent (NOI) to comply with the terms of the General Permit. The prison facilities are identified as SIC 9223, which is not a listed SIC in 40 CFR Section 122.26 (b)(14). An NOI is not required for this facility.

18. Depth to first ground water is approximately 115 feet below ground surface. Regional ground water generally flows southwesterly but may vary seasonally depending on hydraulic influences of nearby supply wells. Ground water is of excellent quality, with electrical conductivities of approximately 300 μmhos/cm.

19. The beneficial uses of underlying ground water are domestic, industrial, and agricultural supply.

20. The upper 3 to 4 feet of site soils consist of moderately dense silty fine sand with some clay. Soils below this depth consist of interbedded layers of dense sandy silt and silty sand to approximately 20 to 40 feet below ground surface.

21. The existing effluent disposal field is fenced and posted on the north and west sides to preclude public access. There is no fence on the east and south sides of the disposal field since they are within prison property. The proposed effluent disposal field will be fenced and posted similar to above with signs and fences located on the south and west boundaries of the field.

22. The California Department of Health Services has established statewide reclamation criteria in Title 22, California Code of Regulations (CCR), Section 60301 et seq. (hereafter Title 22), for use of reclaimed water, and has developed guidelines for specific uses.

23. Madera County adopted a final environmental impact report (EIR) on 21 August 1991 in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. Compliance with these requirements will mitigate or avoid any impacts to water quality.

24. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.

25. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 89-042 is rescinded and the California Department of Corrections, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of waste to surface waters or surface water drainage courses is prohibited.

2. Bypass or overflow of untreated or partially treated waste is prohibited.

3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Title 23, CCR, Section 2510, et. seq. (hereafter Chapter 15), is prohibited.

4. The discharge of waste to irrigation fields from 1 December through February is prohibited.

B. Discharge Specifications:

1. Subject to compliance with Provision F.3, the monthly average dry weather (May through October) discharge shall not exceed 1.454 mgd.

2. Nuisance odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.

3. As a means of discerning compliance with Discharge Specification B.2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.

4. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.

5. The effluent from the treatment facility shall not exceed the following limits:

Constituent	Units	Monthly Average	Daily Maximum
BOD_5[1]	mg/l	40	80
Settleable Solids	ml/l	0.2	1.0

[1] 5-day, 20° Celsius biochemical oxygen demand.

6. The maximum electrical conductivity (EC) of the discharge shall not exceed 900 μmhos/cm.

7. Ponds shall not have a pH less than 6.5 or greater than 9.5.

8. Ponds shall be managed to prevent breeding of mosquitos. In particular:

   a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.

   b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.

   c. Dead algae, vegetation, and debris shall not accumulate on the water surface.

9. Public contact with wastewater and reclamation water shall be precluded through such means as fences, signs, or other acceptable alternatives.

10. Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the nonirrigation season. Design seasonal precipitation shall be based on total annual precipitation using a return period of 25 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically).

11. On or about 1 October of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification B.10.

C. Sludge Disposal:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Chapter 15 and approved by the Executive Officer.

2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer **at least 90 days in advance** of the change.

3. Use and disposal of sewage sludge shall comply with existing federal and state laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.

If the State Water Resources Control Board and the Regional Water Quality Control Boards implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

D. Ground Water Limitations:

The discharge, in combination with other sources, shall not cause underlying ground water to:

1. Contain waste constituents in concentrations statistically greater than receiving water limits, where specified below, or background water quality where not specified. (For purposes of comparison, background water quality shall be determined when background monitoring provides sufficient data. Quality determined in this manner establishes "water quality protection standards.")

2. Contain chemicals, heavy metals, or trace elements in concentrations that adversely affect beneficial uses or exceed maximum contaminant levels specified in CCR, Title 22, Division 4, Chapter 15.

3. Exceed a most probable number of total coliform organisms of 2.2/100 ml over any seven-day period.

4. Exceed concentrations of radionuclides specified in CCR, Title 22, Division 4, Chapter 15.

5. Contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.

6. Contain concentrations of chemical constituents in amounts that adversely affect agricultural use.

E. Reclamation Specifications

1. Use of reclaimed water shall be limited to surface irrigation of fodder, fiber, and seed crops for nonhuman consumption.

2. Grazing of milking animals within the area irrigated with effluent is prohibited.

3. Areas irrigated with reclaimed water shall be managed to prevent breeding of mosquitos. More specifically:

   a. Tail water must be returned and all applied irrigation water must infiltrate completely within a 48-hour period.

   b. Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.

   c. Low-pressure and unpressurized pipelines and ditches accessible to mosquitos shall not be used to store reclaimed water.

4. Reclaimed water used for irrigation shall be managed to minimize erosion, runoff, and movement of aerosols from the disposal area.

5. The Discharger may not irrigate effluent during periods of precipitation and for at least 24 hours after cessation of precipitation.

6. The following setback distances shall be provided at the reclamation site:

Setback Distance, feet	To
25	property line
30	public roads
30	drainage courses
100	on-site irrigation wells
100	food crops
500	domestic wells[1]

[1] For the off-site domestic well identified in Finding No. 14, the setback distance shall not be less than 200 feet.

7. The perimeter of the reclamation area shall be graded to prevent ponding along public roads or other public areas.

8. Application of reclaimed water to the reclamation areas shall be at reasonable rates considering the crop, soil, climate, and irrigation management system.

9. There shall be no standing water in any subarea of the reclamation area 24 hours after each irrigation of that subarea.

10. Storm water runoff from the reclamation area shall not be discharged to any surface water drainage course within 30 days of the last wastewater application.

F. Provisions:

1. The Discharger shall comply with Monitoring and Reporting Program No. 93-235, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.

2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."

3. The monthly average dry weather discharge shall not exceed 0.685 mgd until expansion of the WWTF and disposal facilities is certified as complete by a civil engineer and verified by the Board. Thereafter, the monthly average dry weather flow shall not exceed the limit in Discharge Specification B.1.

4. Prior to use of the expanded WWTF, the Discharger shall submit an updated copy of the operation and maintenance manual reflecting augmentation of the WWTF.

5. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

   To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the State of incorporation if a corporation, the name and address and telephone number of the persons responsible for contact with the Board, and a statement. The statement shall comply with the signatory paragraph of Standard Provision B.3 and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code. Transfer shall be approved or disapproved by the Executive Officer.

6. The Discharger shall use the best practicable cost-effective control technique currently available to comply with salinity limits specified in this order.

7. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

8. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.

9. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes". Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with 22 CCR, Division 4.

10. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 3 December 1993.

William H. Crooks

WILLIAM H. CROOKS, Executive Officer

MRM:cjs: 12/3/93

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 93-235

FOR
CALIFORNIA DEPARTMENT OF CORRECTIONS
PLANNING AND CORRECTIONS DIVISION
FAIRMEAD PRISON COMPLEX
WASTEWATER TREATMENT FACILITY
MADERA COUNTY

Specific sample station locations shall be established with concurrence of the Board's staff, and a description of the stations shall be submitted to Board and attached to this Program.

INFLUENT MONITORING

Influent samples shall be collected at the inlet of the headworks and approximately the same time as effluent samples. Influent monitoring shall include at least the following:

Constituents	Units	Type of Sample	Sampling Frequency
Total Daily Flow	mgd	Continuous	Daily
Settleable Solids	ml/l	grab	Daily
BOD_5[1]	mg/l	8-hr. composite[2]	Weekly
Total Suspended Solids	mg/l	8-hr. composite[2]	Weekly

1 Five-day, 20° Celsius biochemical oxygen demand.

2 Samples to be collected in proportion to flow.

EFFLUENT MONITORING

Except for flow, which may be measured whether at the headworks or treatment unit, effluent samples shall be collected just prior to discharge to the disposal facility. Effluent samples should be representative of the volume and nature of the discharge. Samples collected from the outlet structure of ponds will be considered adequately composited. Time of collection of a grab sample shall be recorded. Effluent monitoring shall include at least the following:

Constituents	Units	Type of Sample	Sampling Frequency[1]
Settleable Solids	ml/l	Grab	Daily
pH	pH Units	Grab	Daily
Dissolved Oxygen[2]	mg/l	Grab	3/week
BOD_5[3]	mg/l	8-hr. composite	Weekly
Total Suspended Solids	mg/l	Grab	Weekly
Electrical Conductivity	µmhos/cm	Grab	Monthly
Kjeldahl Nitrogen (As N)	mg/l	Grab	Quarterly
Nitrates as N	mg/l	Grab	Quarterly

[1] If results of monitoring a pollutant appear to violate effluent limitations, but monitoring frequency is not sufficient to validate violation (e.g., the monthly average for BOD), or indicate a violation and potential upset of the treatment process (e.g., less than minimum D.O.), the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem.

[2] Samples shall be collected at a depth of one foot from each pond, opposite the inlet, and analyzed for dissolved oxygen. Samples shall be collected between 0800 and 0900 hours.

[3] Five-day, 20° Celsius biochemical oxygen demand.

SLUDGE MONITORING

A composite sample of sludge shall be collected annually in accordance with EPA's *POTW Sludge Sampling and analysis guidance document, August 1989,* and tested for the following metals:

Arsenic	Copper	Molybdenum	Zinc
Cadmium	Lead	Nickel	
Chromium	Mercury	Selenium	

Sampling records shall be retained for a minimum of five years. A log shall be kept of sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log should be complete enough to serve as a basis for part of the annual report.

By **3 March 1994,** and annually by **30 January thereafter,** the Discharger shall submit:

a. Annual sludge production in dry tons and percent solids.

b. A schematic diagram showing sludge handling facilities and a solids flow diagram.

c. Depth of application and drying time for sludge drying beds.

d. A description of disposal methods, including the following information related to the disposal methods used at the facility. If more than one method is used, include the percentage of annual sludge production disposed by each method.

   (1) For **landfill disposal,** include (a) the Order numbers of WDRs that regulate the landfill(s) used, (b) the present classifications of the landfill(s) used, and (c) the names and locations of the facilities receiving sludge.

   (2) For **land application,** include (a) the locations of the site(s), (b) the Order numbers of WDRs that regulate the site(s), (c) the application rate in lbs/acre/year (specify wet or dry), and (d) subsequent uses of the land. The Discharger shall comply with 40 CFR 503, *Standards for the Use and Disposal of Sewage Sludge,* should comply with applicable portions of the State Guidance Manual issued by the Department of Health Services, titled *Manual of Good Practice for Landspreading of Sewage Sludge.*

   (3) For **incineration,** include (a) the names and locations of the site(s) where sludge incineration occurs, (b) the Order numbers of WDRs that regulate the sites(s), (c) the disposal method of ash, and (d) the names and locations of facilities receiving ash (if applicable).

   (4) **For composting,** include (a) the location of the site(s), and (b) the Order numbers of WDRs that regulate the site(s).

By **3 March 1994,** the Discharger shall submit characterization of sludge quality, including sludge percent solids and quantitative results of chemical analysis for the priority pollutants listed in 40 CFR 122, Appendix D, Tables II and III (excluding total phenols). All sludge samples shall be a composite of a minimum of twelve (12) discrete samples taken at equal time intervals over 24 hours. Suggested methods for analysis of sludge are provided in EPA publications titled *Test Methods for Evaluating Solid Waste: Physical/Chemical Methods* and *Test Methods for Organic Chemical Analysis of Municipal and Industrial Wastewater.* Recommended analytical holding times for sludge samples should reflect those specified in 40 CFR 136.6.3(3). Other guidance is available in EPA's *POTW Sludge Sampling and Analysis Guidance Document, August 1989.*

Prior to any disposal or land application of sewage sludge, or removal of sewage sludge from the site, the monitoring and recordkeeping requirements of 40 CFR 503 shall be met.

POND MONITORING

The freeboard shall be monitored on the evaporation/percolation ponds in use to the nearest tenth foot. Pond water monitoring shall include the following:

Constituent	Unit	Measurement	Frequency
Freeboard	feet	Observation	Weekly

Permanent markers shall be placed in the pond with calibration indicating the water level at design capacity and available operational freeboard.

In addition, the Discharger shall inspect the condition of the ponds once per week and write visual observation in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the pond (e.g., dark sparkling green, dull green, yellow, grey, tan, brown, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the O&M manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

GROUND WATER MONITORING

By **3 March 1994,** the Discharger shall submit a work plan for a ground water monitoring network with a schedule for implementation to monitor wastewater reuse in all areas where wastewater is stored or reclaimed by the Discharger. The monitoring network shall consist of one or more background monitoring wells and sufficient downgradient wells to determine flow direction and gradient, and to monitor reclamation areas. All well locations and construction features are subject to the prior approval of the Executive Officer and must be sufficient to monitor potential impacts of the disposal operation on the uppermost ground water aquifer. Existing wells proposed for inclusion in the program shall have known construction features (depth, length of perforated interval, surface seal, etc.). **Within 30 days following approval of the work plan by the Executive Officer,** the Discharger shall implement the proposed ground water monitoring well network.

Samples shall be taken monthly from approved background monitoring well(s) for one year and analyzed for the parameters specified below. Data from these analyses shall be reported to the Board **within 30 days after said year ends,** for use in determining water quality protection standards.

If subsequent sampling of the background monitoring well(s) indicates significant water quality changes due to either seasonal fluctuations or other reasons unrelated to waste disposal activities, the Discharger may request modification of the water quality protection standards.

The downgradient wells shall constituent "points of compliance" (POCs). In conjunction with background monitoring, monitoring of POCs will enable one to determine compliance with water quality protection standards. The ground water surface elevation (in feet and hundredths, M.S.L.) in all wells shall be measured. This information shall be displayed on a water flow net diagram for the site. Water samples shall be collected from wells in the approved monitoring network and analyzed as follows:

Constituent	Units	Type of Sample	Sampling Frequency
Standard Minerals[1]	mg/l	Grab	Annually
Electrical Conductivity	µmhos/cm	Grab	Annually
pH	pH Units	Grab	Annually
Total Coliform Organisms	MPN/100ml	Grab	Annually

[1] Standard minerals as used in this program shall include all major cations and anions and include a verification that the analysis is complete (i.e., cation/anion balance).

Following each sampling event (after establishment of water quality protection standards), the Discharger shall determine whether there is statistically significant increase over water quality protection standards for each parameter and constituent analyzed. If the Discharger or the Board finds there is a statistically significant increase in indicator parameters or waste constituents over the water quality protection standards at the POCs, the Discharger shall notify the Board, or acknowledge the Board's findings, and submit, **within 90 days,** either a technical report with a plan and time schedule for implementing a verification monitoring program or a report demonstrating water quality protection standards have been exceeded and assess the horizontal and vertical extent of the impact.

If the Discharger, through a verification monitoring program, or the Board verifies that water quality protection standards have been exceeded at or beyond the POCs, the Discharger shall notify the Board, or acknowledge the Board's findings, and submit a technical report **within 90 days.** The report must contain a plan and time schedule for implementing a corrective action program designed to achieve compliance with water quality protection standards.

ANNUAL LAND MANAGEMENT REPORTS

The Discharger shall submit a summary report on the land management operation after conclusion of each season. The report shall discuss total water application over the season; the total wastewater applied; the total nutrient loading from wastewater, sludges, and chemical fertilizers; and amount of nutrients removed through harvest of the crop. In short, the report shall present a mass balance relative to pollutants of concern and hydraulic loading. The report is due by 30 January of the following year.

REPORTING

Monthly monitoring reports shall be submitted to the Board by the 20th day of the following month.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements, including calculation of all averages, etc.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

By **31 January of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

a. The names, titles, certificate grade (if any) and general responsibilities of persons operating and maintaining the WWTF.

b. The names and telephone numbers of persons to contact regarding the WWTF for emergency and routine situations.

c. A certified statement of when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who did the calibration (Standard Provision C.4).

d. A statement whether the current operation and maintenance manual, and contingency plan, reflect the WWTF as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.

e. The total quantity of sludge disposed of during the previous year and ultimate disposal site(s).

All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

The Discharger shall implement the above monitoring program on the first day of the month following adoption of this Order.

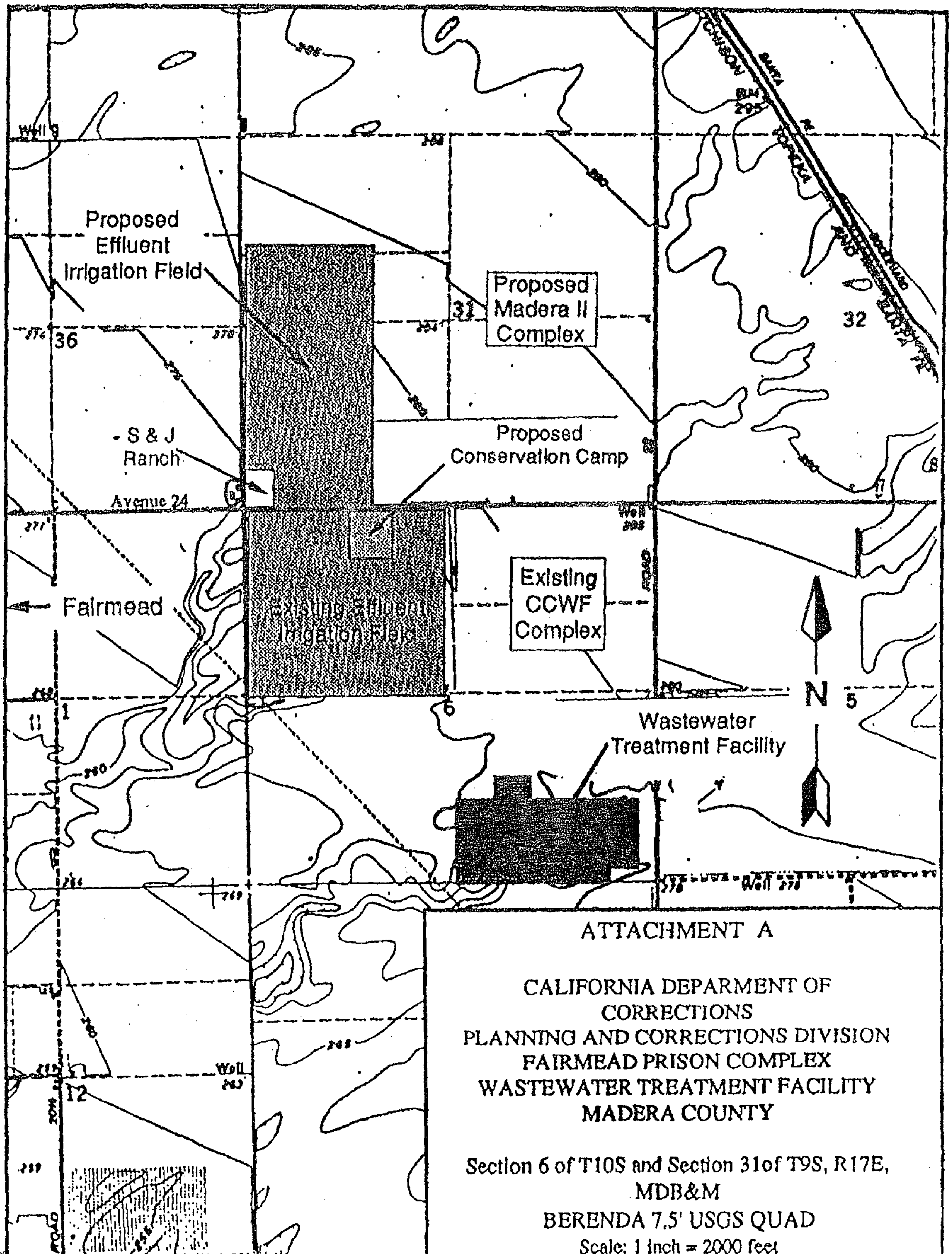
Ordered by: William H. Crooks

WILLIAM H. CROOKS, Executive Officer

3 December 1993

(Date)

MRM:cjs: 12/3/93



ATTACHMENT A

CALIFORNIA DEPARMENT OF CORRECTIONS
PLANNING AND CORRECTIONS DIVISION
FAIRMEAD PRISON COMPLEX
WASTEWATER TREATMENT FACILITY
MADERA COUNTY

Section 6 of T10S and Section 31of T9S, R17E, MDB&M
BERENDA 7.5' USGS QUAD
Scale: 1 inch = 2000 feet

INFORMATION SHEET

CALIFORNIA DEPARTMENT OF CORRECTIONS
FAIRMEAD PRISON COMPLEX
WASTEWATER TREATMENT FACILITY
MADERA COUNTY

The California Department of Corrections (Discharger) operates a wastewater treatment facility (WWTF) approximately 6 miles southeast of Chowchilla and 12 miles northwest of Madera, in Madera County. The WWTF is regulated by Waste Discharge Requirements Order No. 89-042.

The existing prison facility, known as the Central California Women's facility (CCWF), has a capacity of 2,000 beds but can reach 190% of capacity, or 3,800 beds. The facility generates an average of 0.685 mgd of domestic wastewater, the current design capacity of the WWTF.

The existing WWTF provides pretreatment with a grease trap, a mechanically cleaned bar screen, and a grinder. Following pretreatment, the wastewater is treated in two 3.1-acre aerated lagoons followed by two 3.1-acre facultative ponds. Wastewater is then discharged to four evaporation/percolation ponds prior to reclamation on 124 acres of alfalfa. The reclamation areas are developed to operate with a tail water collection and return system to mitigate problems associated with wastewater ponding and runoff. Sludge removed from the wastewater ponds is first dried, then used as a soil amendment at locations designated for wastewater reclamation.

The Discharger is proposing to build a second prison nearby with approximately the same capacity as CCWF, and a minimum security prison camp just northwest of CCWF. Wastewater from both prisons will gravity flow to the existing WWTF where average flows will increase to 1.454 mgd.

The Dischargers plans to expand the WWTF by installing an additional bar screen, grinder, and an aerated lagoon and facultative pond. The wastewater disposal facilities will also be augmented with three additional evaporation/percolation ponds and an additional 150 acres of agricultural fields.

Surface water drainage at the WWTF is to the San Joaquin River. The beneficial uses of the San Joaquin River (Sack Dam to Merced River) are agricultural and industrial supply, recreation, wildlife habitat, fresh water habitat, fish migration and spawning.

A portion of the proposed effluent disposal field lies within the 100-year floodplain. Measures will be taken to ensure contact between reclamation water and surface water runoff from precipitation is minimized. Wastewater will not be reclaimed during months with the highest rainfall (December through February).